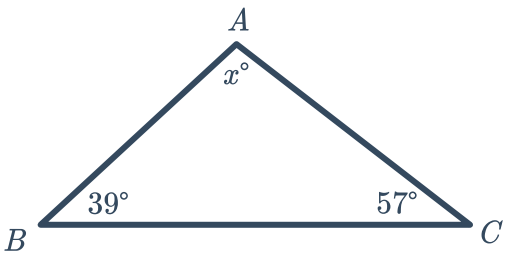
# Angles of triangles

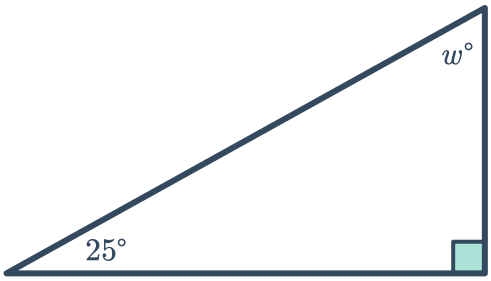
**1** State the sum of the measure of the interior angles of any triangle.

**2** For the following diagrams, solve for the unknown.

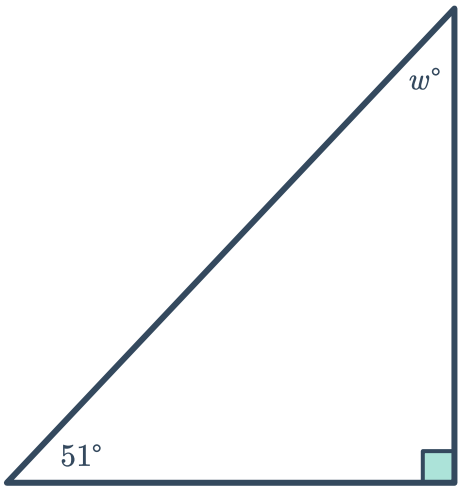
**a**



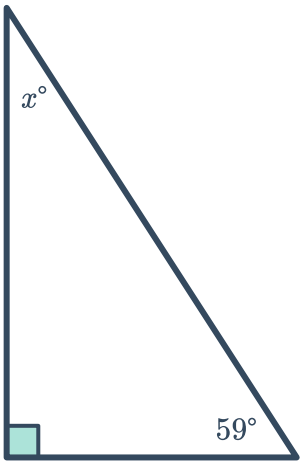

**b**



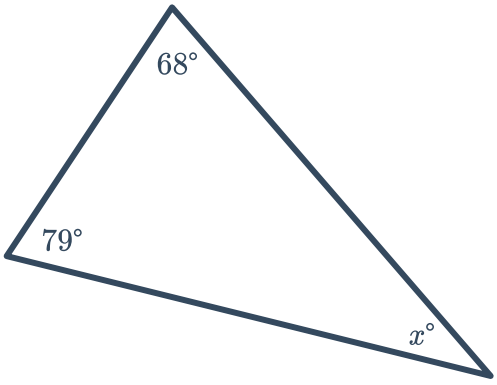

**c**




**d**

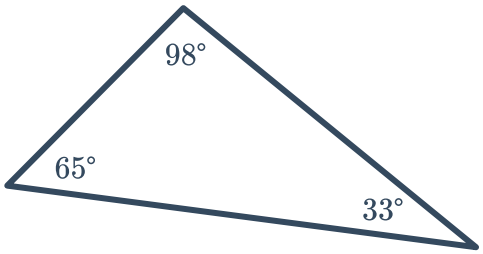



**e**




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**3** State the reason why it is not possible to make the following triangle.




**4** Find the measure of the third angle of a triangle whose two angles measure $72^\circ$ and $58^\circ$.

**5** Classify the triangle that has two angles which measure $15^\circ$ and $150^\circ$.

**6** Consider a triangle whose three angles measure $x^\circ$, $y^\circ$ and $48^\circ$.
Determine whether the following statements are correct.

**a** $x + y = 48$

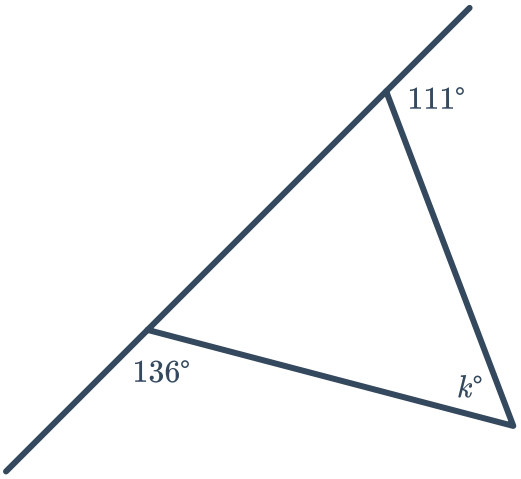
**b** $y = x + 48$

**c** $y + x = 180 - 48$

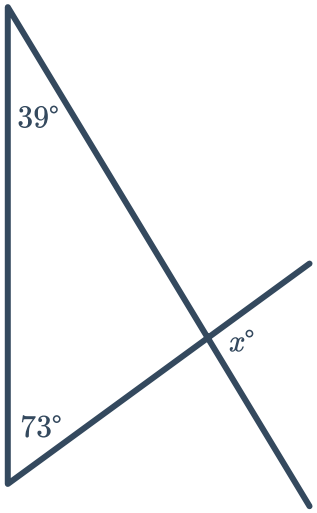
**d** $y = 180 - x$

**7** For the following diagrams, solve for the value of the variable.

**a**

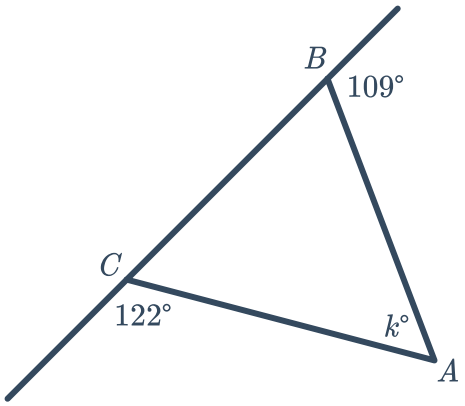



**b**

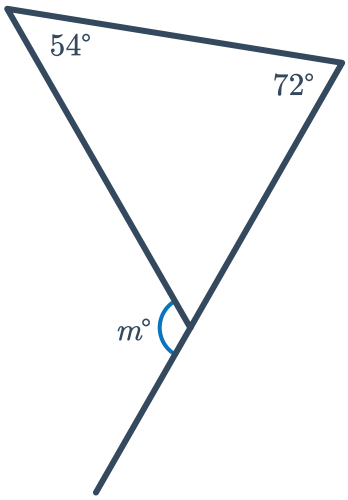



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**c**

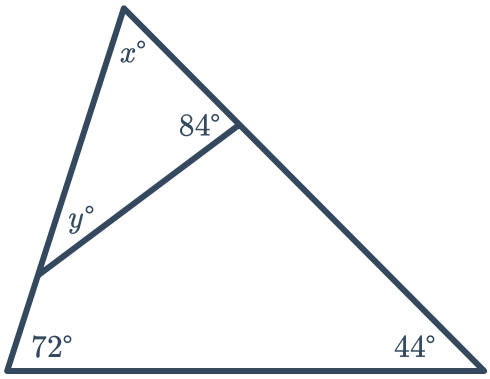



**d**

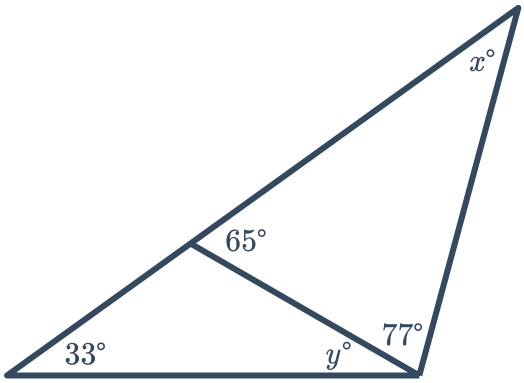



**8** For the following diagrams, solve for the values of $x$ and $y$.

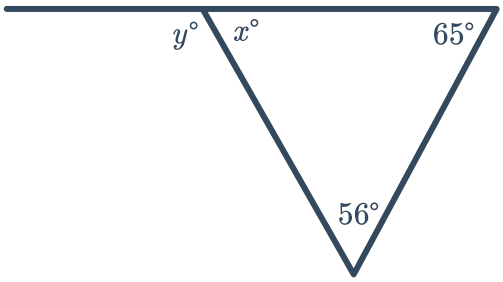
**a**



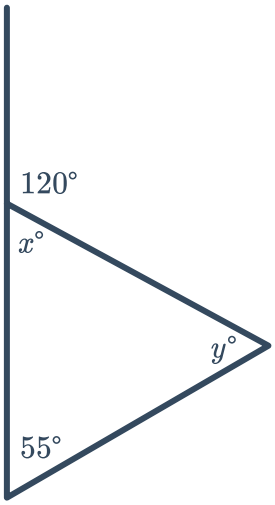

**b**




**c**




**d**

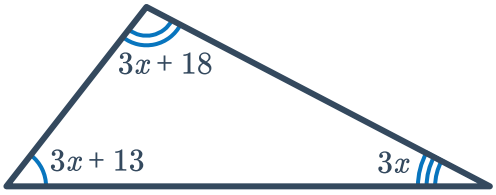



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**9** Consider the given diagram:

**a** Solve for the exact value of $x$.

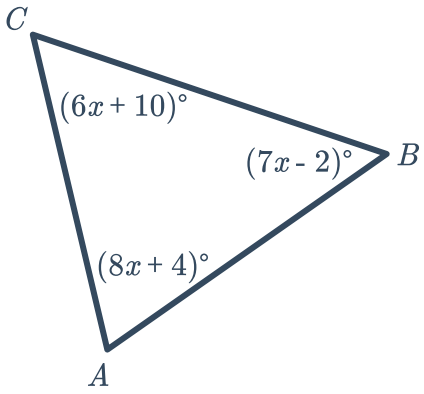
**b** Hence, find the exact value of the smallest angle in the triangle.




**10** Consider the given diagram:

**a** Solve for $x$.

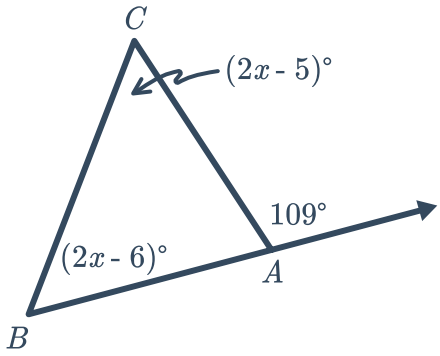
**b** Hence, find $m\angle BAC$.




**11** Consider the given diagram:

**a** Solve for $x$.

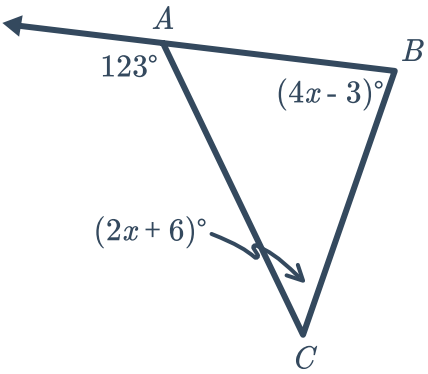
**b** Hence, find $m\angle ABC$.




**12** Consider the given diagram:

**a** Solve for $x$.

**b** Hence, find $m\angle ACB$.



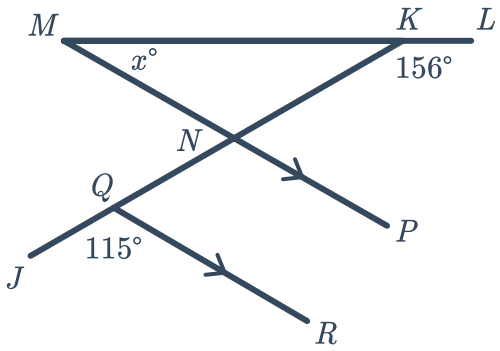

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**13** Consider the given diagram:

**a** Describe the relationship of $\angle JQR$ and $\angle JNP$.

**b** Describe the relationship of $\angle MNK$ and $\angle JNP$.

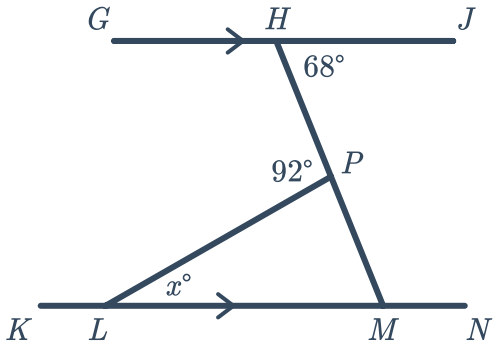
**c** Hence, solve for $x$.




**14** Consider the given diagram:

**a** Describe the relationship of $\angle JHM$ and $\angle KMH$.

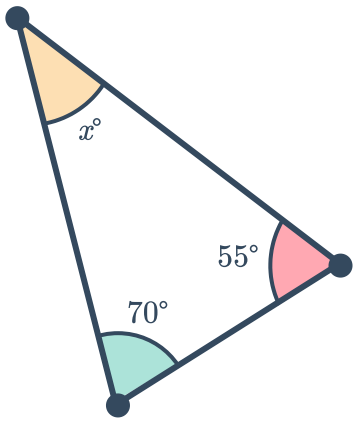
**b** Hence, solve for $x$.



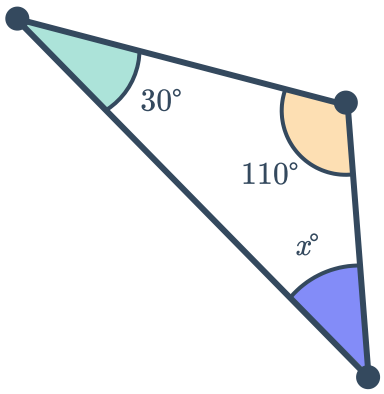

## Additional questions

**15** Find the value of $x$ in the following diagrams:

**a**

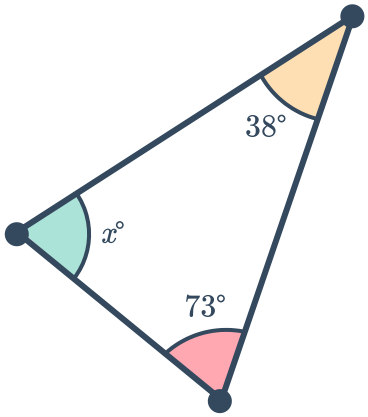



**b**

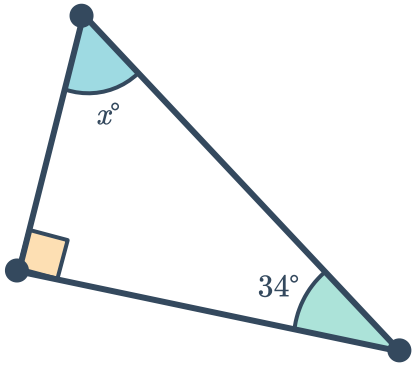



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**c**




**d**




**16** Find the measure of the third angle of a triangle given the measures of the other two angles:

**a** $36^\circ$ and $94^\circ$ **b** $45^\circ$ and $45^\circ$ **c** $3^\circ$ and $150^\circ$ **d** $75^\circ$ and $x^\circ$

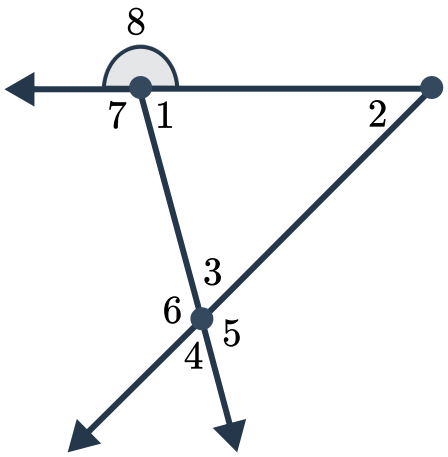
**17**

**a** Identify the numbered angles that are exterior angles.

**b** $\angle 7$ is an exterior angle for a specific interior angle. Identify the described interior angle.

**c** Describe how exterior angles $5$ and $6$ are related.

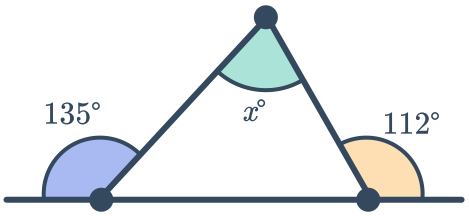
**d** Identify the two interior angle measures that have the same measure as angle $5$? Explain your reasoning.



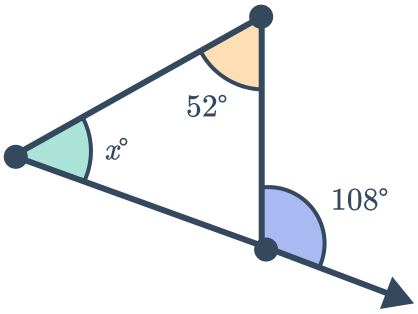

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**18** Find the value of $x$ in the following diagrams:

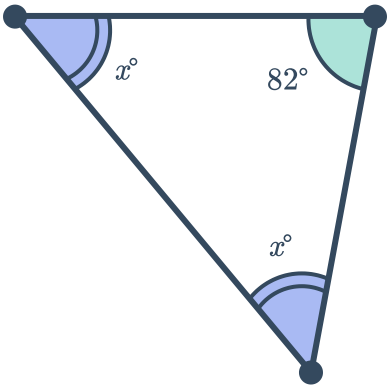
**a**



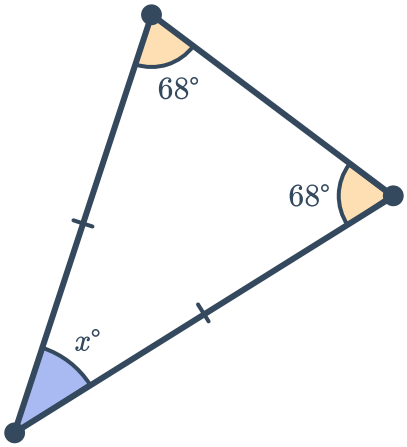

**b**




**c**



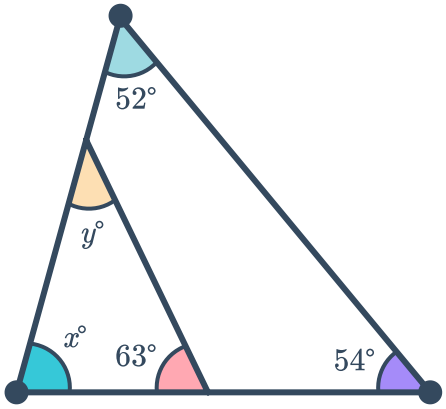

**d**



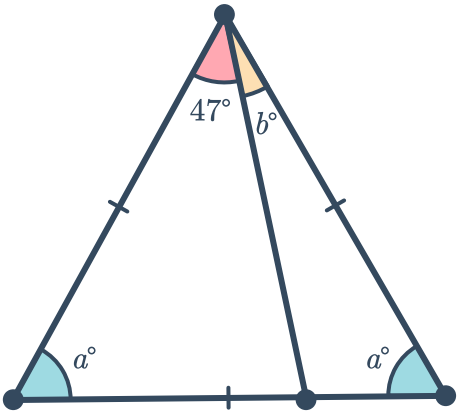

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**19** For the following diagrams, find the value of each variable:

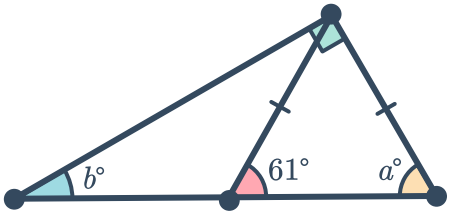
**a**



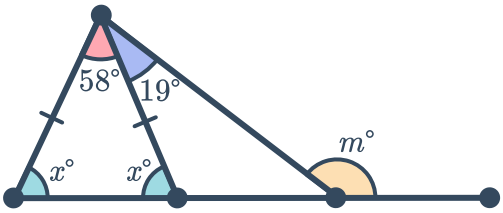

**b**




**c**




**d**




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**20** For each of the below diagrams:

i Find the value of $x$.

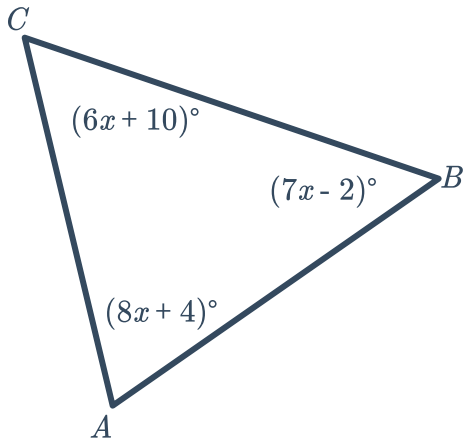
ii Find the measure of $\angle CAB$.

iii Find the measure of $\angle ABC$.

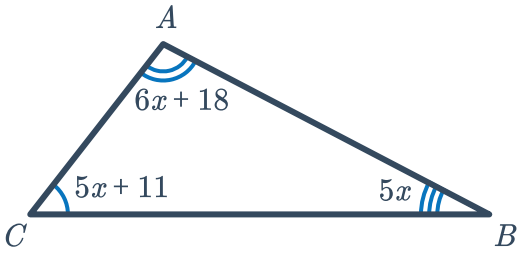
iv Find the measure of $\angle BCA$.

Consider the given triangles and round answers to two decimal places when applicable.

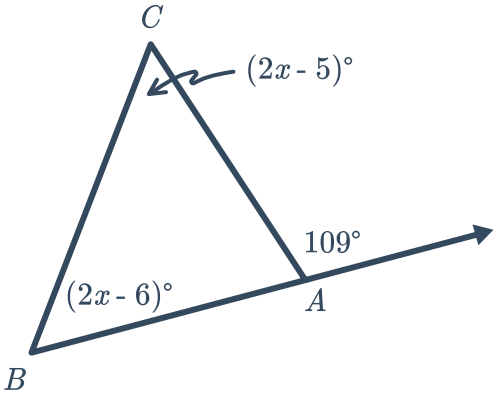
**a**




**b**




**c**




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**21** Determine whether the following statements are true or false. If false, provide a counterexample and correct the statement so it is true. If true, explain why.

**a** The interior angles of only right triangles sum to $180^\circ$.

**b** The measure of an exterior angle of a triangle is equal to the sum of the two remote interior angles of the triangle.

**c** The sum of the measures of the exterior angles of a triangle is $180^\circ$.

**d** In a right triangle, the right angle is $90^\circ$ and the remaining two angles are always congruent.

**22** Miriam has an idea that she shares with her math class. Miriam's claim is that each pair of remote interior angles in a triangle has two exterior angles. Is this correct? Provide a diagram to support your answer.

**23** A triangle has two angles with measures of $37$ and $85$. What is the measure of the smallest exterior angle of the triangle. Explain your reasoning.

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